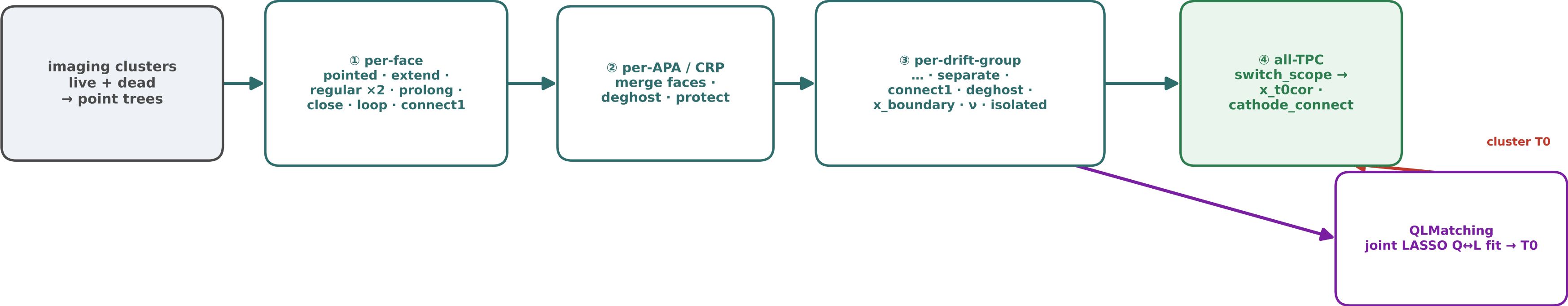


ProtoDUNE HD + VD — Wire-Cell 3-D Clustering + Q/L Matching

one 4-stage MultiAlgBlobClustering cascade + joint QLMatching T0 pinning, two detector geometries / photon models



	① multiplicity	② multiplicity	③ drift groups	photon detectors	photon model
PD-HD	×4 (1 imaging face / APA)	×4 APAs	{APA0,2}·face0 / {APA1,3}·face1	160 flat X-ARAPUCA (ch 0–79 +x · 80–159 −x)	semi-analytical VUV · QtoL 1.0
PD-VD	×16 (8 anodes × 2 faces)	×8 CRP anodes	{anodes 0–3} / {anodes 4–7}	40 PDs: cathode / membrane XA + PMT families	library 128 nm Ar · QtoL 0.094

PD-HD: [cfg/.../pdhd/{clus,qlmatching}.jsonnet](#) · see [pdhd/docs/clustering-qlmatching-chain-diagram.md](#)

PD-VD: [cfg/.../protodunevd/{clus,qlmatching}.jsonnet](#) · see [pdvd/docs/17_pdvd-clustering-qlmatching-chain.md](#)